KEERTHISHREE KESAVAN

DOMAIN: Machine Learning | Full-Stack | Data Science
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EDUCATION

Bachelor of Technology in Computer Science Engineering (Artificial Intelligence & Machine Learning) Aug 2023 - Present
Sri Ramachandra Institute of Higher Education and Research Chennai, India
CGPA: 9.0/10 and Current GPA: 9.8/10

Vel Tech Dr. RR & DR. SR Matriculation Higher Secondary School Year of Completion - 2023
Class XII (State Board) percentage: 87% Chennai, India

TECHNICAL SKILLS

Machine Learning: Scikit-learn, AutoML, Reinforcement Learning, SHAP, LIME (XAI)
Deep Learning and LLMs: Llama-3, BGE Embeddings (HuggingFace), Policy Gradient , CNN, NLP
Web Development: React.js, Node.js, Express.js, REST APIs, Socket.io, BullMQ, JWT, HTML, CSS, JavaScript
Data & Visualization: SQL, Tableau, Plotly, Streamlit, Pandas, NumPy
Cloud & DevOps: AWS (Lambda, S3, DynamoDB, Textract, Rekognition, Transcribe), Docker
Databases: MongoDB, MySQL, Redis

INTERNSHIP

Machine Learning and Data Science Intern - REUDE Technologies, Chennai May 2025 - July 2025
Tech Stack: Python, Scikit-learn, Machine Learning, Streamlit, Open3D, Data Visualization

- Built predictive ML models to detect patterns, anomalies in spatial datasets with strong validation performance.
- Designed Python-based feature engineering and preprocessing pipelines to deploy insights via a Streamlit dashboard.
- **Impact:** Reduced manual analysis effort and improved interpretability of spatial data for faster decision-making.

RESEARCH AND PUBLICATION

Hybrid SVM-Naïve Bayes Ensemble with LIME and SHAP for Transparent Air Quality and Health Risk Prediction
13th IEEE International Conference on Intelligent Systems and Embedded Design (ISED-2025), NIT Raipur

- Co-authored and presented an IEEE conference paper on a hybrid ensemble model for AQI and multi-disease health risk.
- Developed a hybrid SVM–Gaussian Naive Bayes model with **96.2% AQI classification accuracy** and strong F1-score.
- Integrated LIME and SHAP for local and global explainability, ensuring transparent and trustworthy predictions.
- Built an interactive geospatial dashboard for real-time air quality and health risk visualization for smart city deployment.

PROJECTS

ReviewGuard - AI-Powered Review Moderation Platform | React, Node.js, Express.js, MongoDB Atlas, Redis, BullMQ, Socket.io, Llama-3 (Groq), BGE (BAAI) Embeddings(Hugging face) [**Live Web** | **GitHub**]

- Architected a multi-layered AI detection engine using **Llama-3** (Groq) and **BGE (BAAI) embeddings** (Hugging Face) to detect toxic, semantic duplicate, and overlapping violations (content flagged as both toxic and redundant).
- Built a scalable async pipeline with BullMQ and Redis, reducing API latency by **~90–95% (1.2s → <100ms).**
- Developed a **real-time** system with **Socket.io** for instant user feedback and live admin threat monitoring.
- Engineered a Hybrid Defense Mechanism **(LLM + BGE + TF-IDF/Regex fallback)** for reliability during AI/network failures.
- Secured RBAC platform (Admin, Moderator, User) with JWT authentication, Bcrypt password hashing, email verification, password reset workflows, audit logs, and rate limiting.

Automated Descriptive Answer Evaluation System | Python, BERT, NLP, AWS Textract, Streamlit

- Built an AI system to auto-score subjective answers using **BERT**-based semantic similarity, solving inconsistent manual grading by evaluating conceptual correctness beyond keyword matching.
- Integrated **AWS Textract** for **OCR extraction** from scanned answer sheets with full **NLP** preprocessing pipeline (tokenization, stop word removal, lowercasing).
- Captures conceptual meaning beyond keyword matching, **outperforming** traditional **TF-IDF** approaches.

3D RL-Based Traffic Signal Control System with Ambulance Priority | Python, Reinforcement Learning, Policy Gradient, Q-Learning, Matplotlib, Seaborn, 3D Simulation

- Designed an intelligent traffic management system using Reinforcement Learning **(Policy Gradient & Q-Learning)** to optimize signal timing and reduce congestion.
- Developed a real-time 3D simulation environment to monitor traffic flow; achieved **45% reduction** in traffic wait times.
- Implemented ambulance priority logic resulting in **83% faster** emergency response times and built a live dashboard to visualize RL reward trends and agent performance.

Heart Disease Risk Prediction using AutoML and Explainable AI | Python, AutoML, Streamlit, SHAP (XAI)

- Built an AutoML system predicting heart disease risk, achieving **~76% validation accuracy** with personalized risk scores.
- Applied SHAP-based explainability technique to provide feature-level interpretability, enabling transparent assessment.
- Deployed via Streamlit with interactive visualizations and downloadable PDF/CSV reports for generated results.

Explainable AI for Air Quality & Health Risk Prediction using Hybrid ML Model | Python, SVM, Naive Bayes, SHAP, LIME, Geospatial Visualization

- Built a Hybrid ML model **(SVM + GNB)** achieving **96.2% AQI prediction accuracy** with SHAP/LIME-based explainability for transparent model insights.
- Designed an AI-powered interactive geospatial map for real-time air quality and health risk monitoring and visualization.

AutoVerify – AWS Powered Driving License Verification System | AWS (Textract, Rekognition, Lambda, S3, DynamoDB), Streamlit, Python

- Built a serverless AWS-based license pipeline (Textract → Lambda → DynamoDB → S3) to automate driving license data extraction, eliminating manual entry errors.
- Achieved high-accuracy structured data validation with real-time retrieval and visualization via a Streamlit frontend.

ACHIEVEMENTS AND INVOLVEMENTS

Soft Skill Development – NPTEL (8-week course, 2026): Achieved 90% (Elite + Gold); ranked in Topper (Top 1%)

Effective Writing – NPTEL (8-week course, 2026): Secured 87% (Elite + Silver); ranked in Topper (Top 5%)

IEEE Publication (ISED-2025): Co-authored and presented research at NIT Raipur on hybrid ML for AQI prediction

AWS Academy Machine Learning Foundations (**Digital Badge**) | **IBM Z Day 2025 – IBM Z Skills** (**Digital Badge**)

MongoDB and Linux Course Completion: Completed and Certified in MongoDB and Linux system administration.

Hackathon and Tech Expo Participation: Shortlisted in a hackathon, showcased projects at Tech Expo and Research Day.

Industrial Visit (Feb 2025): Attended an industrial visit to Tidel Park, gaining exposure to real-world industry practices.

LEADERSHIP POSITIONS

Research Club Coordinator (2023 – 2027):

- Hosted various research club and technical events(Tech Expo, Research Day, and Mindspark) and conducted engaging and informative activities for students, managing coordination, planning and scheduling.

NCC Certified (Lance Corporal – LCpl): 13 (TN) BN NCC, Government of India

- Achieved NCC Certificate ‘A’ (Grade A) under National Cadet Corps, Government of India. Trained in disciple, leadership and drill activities.